

APPLIED MATHEMATICS · RESEARCH PORTFOLIO

Deep Research Foundations

*Principles, Quotes & Mathematical Foundations
Across Quantitative Finance, Data Science, AI Engineering & Quantum
Computing*

Quantitative Trading • Data Science •
AI Engineering • Quantum Computing

Twelve rigorous projects · Nobel Prize & Turing Award context
Compiled August 2026

Table of Contents

Part I --- Quantitative Trading & Stochastic Control

- 1.1 Optimal Execution via the Almgren–Chriss Framework
- 1.2 Markowitz Portfolio Optimization & Modern Extensions
- 1.3 Stochastic Volatility & the Heston Model

Part II --- Data Science & Advanced Statistics

- 2.1 Bayesian Nonparametrics: Dirichlet Process Mixtures
- 2.2 Causal Inference: Double Machine Learning
- 2.3 High-Dimensional Statistics: Compressed Sensing

Part III --- AI Engineering & LLM Systems

- 3.1 Retrieval-Augmented Generation (RAG)
- 3.2 Multi-Agent Orchestration with Game Theory
- 3.3 Fine-Tuning with Convergence Analysis (LoRA)

Part IV --- Quantum Computing & Algorithms

- 4.1 Variational Quantum Eigensolver (VQE)
- 4.2 Quantum Approximate Optimization Algorithm (QAOA)
- 4.3 Quantum Error Correction & Surface Codes

Summary & References

Unifying Principles Across All Domains
Complete Bibliography

I QUANTITATIVE TRADING & STOCHASTIC CONTROL

1.1 Optimal Execution via the Almgren--Chriss Framework

HISTORICAL CONTEXT & NOBEL RECOGNITION

NOBEL PRIZE IN ECONOMICS · 1997

The foundation of modern quantitative finance rests on the work of **Robert C. Merton** and **Myron Scholes**, awarded the Nobel Prize for their Black–Scholes option pricing model and the continuous-time framework it established.

“The problem is to find the optimal policy for purchasing or selling a large block of stock over a specified time horizon.”
— **Almgren & Chriss (2000)**

“The investor’s goal is to maximize expected utility of consumption over his lifetime. The dynamic nature of the problem requires continuous rebalancing in response to price changes.”
— **Robert C. Merton, on Dynamic Portfolio Theory (1971)**

CORE PRINCIPLE: THE HAMILTON--JACOBI--BELLMAN EQUATION

The optimal trading problem reduces to solving the HJB equation:

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$$

where $V(S, t)$ is the value function. Almgren–Chriss extends this to incorporate market impact costs.

Temporary Impact (cost per share):

$$\text{Cost} = \lambda \cdot q^\alpha$$

where λ is market depth, q is order size, and $\alpha \in [1.5, 2]$ is the impact elasticity.

Permanent Impact (shift in equilibrium price):

$$\Delta P = \gamma \cdot v$$

where v is the trading rate (shares per unit time).

KEY PAPERS

- Almgren, R., & Chriss, N. (2000). *Optimal Execution of Portfolio Transactions*.
- Gatheral, J. (2010). *No-Dynamic-Arbitrage and Market Impact*.

1.2 Markowitz Portfolio Optimization & Modern Extensions

NOBEL PRIZE IN ECONOMICS · 1990

“The investor considers expected return a desirable thing and variance of return an undesirable thing. The investor should maximize expected return subject to an upper limit on variance.”
 — **Harry Markowitz**

THE MEAN-VARIANCE PROBLEM

Minimize portfolio variance subject to an expected return constraint:

$$\min_{\mathbf{w}} \quad \mathbf{w}^T \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^T \boldsymbol{\mu} = \mu_p, \quad \sum_i w_i = 1$$

- $\mathbf{w}$ — portfolio weights
- Σ — covariance matrix (estimated from returns)
- $\boldsymbol{\mu}$ — expected returns vector

THE PROBLEM: ESTIMATION ERROR

Ledoit & Wolf (2004) showed the sample covariance matrix is **severely ill-conditioned** in high dimensions:

“The sample covariance matrix is known to perform poorly when the number of assets is large relative to the length of the time series. Shrinkage toward the identity matrix dramatically improves performance.”
 — **Ledoit & Wolf (2004)**

Shrinkage Estimator:

$$\Sigma_{\text{shrink}} = \alpha \Sigma_{\text{sample}} + (1 - \alpha) \Sigma_{\text{target}}$$

α is chosen to minimize the Frobenius norm of expected estimation error. Typical targets: single-factor model $\Sigma_{\text{target}} = \beta \beta^T + \text{diag}(\sigma_\epsilon^2)$, or constant correlation $\Sigma_{\text{target}} = \bar{\rho}[\mathbf{1}\mathbf{1}^T - I] + \text{diag}(\sigma^2)$.

BLACK–LITTERMAN EXTENSION

“Investors typically have views about the expected return of certain assets and want to construct portfolios that take advantage of these views.”
 — **Fischer Black & Robert Litterman (1991)**

$$E[\mathbf{r} \mid \text{views}] = \Sigma \mathbf{P}^T [\mathbf{P} \Sigma \mathbf{P}^T + \Omega]^{-1} (\mathbf{Q} - \mathbf{P} \boldsymbol{\mu}_{\text{prior}})$$

$\mathbf{P}$ — view matrix; $\mathbf{Q}$ — view returns; Ω — view confidence (uncertainty of investor views).

1.3 Stochastic Volatility & the Heston Model

“A closed-form solution for European option prices when the volatility of the underlying stock price is stochastic appears to be intractable. Instead, I derive a semi-closed-form solution using characteristic functions.”
 — **Steven Heston (1993)**

THE MODEL

$$dS = \mu S dt + \sqrt{v} S dW_S$$

$$dv = \kappa(\theta - v) dt + \sigma_v \sqrt{v} dW_v$$

- v — instantaneous variance (latent state)
- κ — mean-reversion speed
- θ — long-run variance
- σ_v — volatility of volatility
- ρ — correlation between asset and volatility shocks

CALIBRATION VIA CARR--MADAN (FFT)

Carr & Madan (1999) express the call option price via the characteristic function:

$$C(K, T) = \frac{e^{-rT}}{\pi} \int_0^\infty \frac{e^{-i\nu \log K} \phi(\nu - i/2)}{\nu^2 + 1/4} d\nu$$

where $\phi(\nu)$ is the characteristic function of log-returns. This is vastly faster than Monte Carlo simulation for calibration.

REFERENCE

- Carr, P., & Madan, D. (1999). *Option Valuation Using the Fast Fourier Transform*.

II DATA SCIENCE & ADVANCED STATISTICS

2.1 Bayesian Nonparametrics: Dirichlet Process Mixtures

“The Dirichlet Process provides a convenient mechanism for specifying a prior distribution over the space of probability measures. It is nonparametric in the sense that the prior does not assume a fixed parametric form for the data distribution.”

— **Thomas S. Ferguson (1973)**

THE MODEL

Data $y_1, \dots, y_n \sim \text{Mixture}$, where the number of components K is unknown, component parameters ϕ_k are random, and mixture weights π_k arise from a Dirichlet Process:

$$G \sim \text{DP}(\alpha, G_0) \quad \phi_i \mid G \sim G, \quad y_i \mid \phi_i \sim F(\cdot \mid \phi_i)$$

- α — concentration parameter (higher $\rightarrow$ more clusters)
- G_0 — base measure (prior on component parameters)
- F — likelihood (e.g., Gaussian)

STICK-BREAKING CONSTRUCTION (SETHURAMAN, 1994)

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k} \quad \text{where} \quad \pi_k = V_k \prod_{j < k} (1 - V_j), \quad V_k \sim \text{Beta}(1, \alpha)$$

CHINESE RESTAURANT PROCESS

“Customers enter a restaurant and sit at tables. Customer n sits at an occupied table k with probability $n_k / (n - 1 + \alpha)$, or at a new table with probability $\alpha / (n - 1 + \alpha)$.”

Pólya urn interpretation: clustering naturally emerges from exchangeability — no need to pre-specify the number of clusters.

2.2 Causal Inference: Double Machine Learning

“The fundamental problem of causal inference is that we can only observe one of two potential outcomes for each unit. We never see what would have happened in the counterfactual world.”

— **Donald Rubin, Potential Outcomes Framework (1974)**

THE PROBLEM

Estimate treatment effect $\tau = E[Y(1) - Y(0)]$ in high dimensions where $p \gg n$. Standard regression **fails**: $Y = \beta_0 + \beta_1 T + \beta_p X + \epsilon$, with LASSO biasing the treatment coefficient.

DOUBLE MACHINE LEARNING SOLUTION

“Machine learning methods can estimate nuisance parameters (confounders) without bias to the treatment coefficient if we use Neyman-orthogonal scores.”
— Chernozhukov et al. (2018)

Algorithm:

1. Regress out confounding: $\hat{m}(X) = \text{ML}(Y \sim X)$ and $\hat{\tau}(X) = \text{ML}(T \sim X)$
2. Compute residuals: $\tilde{Y} = Y - \hat{m}(X)$, $\tilde{T} = T - \hat{\tau}(X)$
3. Estimate treatment effect: $\hat{\tau} = \frac{\sum_i \tilde{T}_i \tilde{Y}_i}{\sum_i \tilde{T}_i^2}$

This works because first-stage ML errors do not propagate to the τ estimate, due to Neyman orthogonality:

$$\mathbb{E}[\partial_\nu \psi(Z, \eta_0)] = 0$$

KEY PAPERS

- Chernozhukov, V., et al. (2018). *Double Machine Learning for Treatment and Structural Parameters*.
- Athey, S., & Wager, S. (2019). *Generalized Random Forests*.

2.3 High-Dimensional Statistics: Compressed Sensing

“Compressed sensing shows that one can recover high-dimensional sparse signals from far fewer measurements than the signal dimension, provided the measurement matrix satisfies the Restricted Isometry Property.”
— David Donoho & Emmanuel Candès

THE PROBLEM

Recover $\mathbf{x} \in \mathbb{R}^p$ (sparse, $\|\mathbf{x}\|_0 \leq k$) from $\mathbf{y} = A\mathbf{x} + \epsilon$, where $m \ll p$ (severely underdetermined).

THE SOLUTION: ℓ_1 MINIMIZATION (LASSO)

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{y} - A\mathbf{x}\|_2^2 + \lambda \|\mathbf{x}\|_1$$

THEORETICAL GUARANTEE --- THE RIP CONDITION

If A satisfies the Restricted Isometry Property with constant $\delta_k < 1$:

$$(1 - \delta_k) \|\mathbf{x}\|_2^2 \leq \|A\mathbf{x}\|_2^2 \leq (1 + \delta_k) \|\mathbf{x}\|_2^2$$

for all k -sparse $\mathbf{x}$, then $\hat{\mathbf{x}}$ recovers $\mathbf{x}$ exactly (or with error $O(\epsilon)$). Random Gaussian, Bernoulli, and subsampled Fourier matrices satisfy RIP with high probability.

Measurement Bound:

$$m = O(k \log(p/k))$$

Exponentially better than the naive p measurements — the foundation of modern sparse signal recovery, MRI, and imaging.

III AI ENGINEERING & LLM SYSTEMS

3.1 Retrieval-Augmented Generation (RAG)

“We show how to augment parametric knowledge in a pre-trained sequence-to-sequence model with non-parametric memory provided by a dense vector index of Wikipedia. This retrieval-augmented generation approach sets new state-of-the-art on open-domain QA.”

— **Patrick Lewis et al., Facebook AI Research (2020)**

THE ARCHITECTURE

Question → [Retriever] → Top-K Documents → [Reranker] → Context → [LLM] + Context → Answer

Dense Passage Retrieval:

$$s_i = \frac{q_{\text{emb}} \cdot d_{i,\text{emb}}}{\|q_{\text{emb}}\| \|d_{i,\text{emb}}\|}$$

Cosine similarity between query embedding and document embedding.

INFORMATION-THEORETIC VIEW

Shannon’s channel capacity: $C = \log_2(1 + \text{SNR})$ bits/dimension. RAG performance depends on (1) retrieval precision — probability the relevant document is in the top- K ; (2) context length capacity — attention bandwidth $O(n^2)$; and (3) the information bottleneck balancing coverage against precision in reranking.

3.2 Multi-Agent Orchestration with Game Theory

NOBEL PRIZE IN ECONOMICS · 1994

“It is shown that finite games always have at least one equilibrium point, in mixed strategies. This equilibrium point is the point of rest to which the game outcomes converge in finite games.”

— **John Nash**

AGENT COORDINATION PROBLEM

Multiple agents allocate shared resources (compute, memory, context window). Agent i has strategy set S_i and payoff $u_i(s_1, \dots, s_n)$ depending on all agents’ choices — the goal is to find the Nash equilibrium.

MECHANISM DESIGN (VICKREY--CLARKE--GROVES)

NOBEL PRIZE IN ECONOMICS · 1996

“The Vickrey auction — where bidders bid truthfully at equilibrium — can be generalized to allocate any divisible resource efficiently.”

— **William Vickrey**

VCG payment for agent i :

$$p_i = \sum_{j \neq i} v_j(x^{-i}) - \sum_{j \neq i} v_j(x^*)$$

Agent i pays the external cost imposed on others — truthful bidding is the dominant strategy.

3.3 Fine-Tuning with Convergence Analysis (LoRA)

“We show that fine-tuned models have a low intrinsic dimension — we can update only a small number of rank- r matrices $\Delta W = BA$ rather than the full weight matrix W .”
 — **Edward Hu et al., Microsoft (2021)**

$$W_{\text{adapted}} = W + \Delta W = W + BA$$

$B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times d}$, $r \ll d$ — computational savings: $O(dr)$ vs $O(d^2)$ memory.

CONVERGENCE ANALYSIS

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n -\log p_{\theta}(y_i \mid x_i)$$

Supervised fine-tuning loss, minimized via $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$.

Convergence rate (convex approximation): $L(\theta_T) - L(\theta^*) = O(1/T)$. In practice (non-convex): $\|\nabla L(\theta_T)\|^2 = O(1/T)$, with generalization gap $O(\sqrt{d \log T/n})$.

IV QUANTUM COMPUTING & ALGORITHMS

4.1 Variational Quantum Eigensolver (VQE)

NOBEL PRIZE IN PHYSICS · 2025

“The discovery of macroscopic quantum tunneling and energy quantization in electrical circuits opened the path to superconducting qubits, which form the basis of today’s quantum computers.”
— **John Clarke, Michel Devoret & John Martinis**

THE PROBLEM

Find the ground state energy of a quantum Hamiltonian:

$$E_0 = \min_{\psi} \langle \psi | H | \psi \rangle$$

Classical bottleneck: exponential complexity — state space is 2^n for n qubits.

VQE SOLUTION (PERUZZO ET AL., 2014)

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

Algorithm: (1) initialize ansatz parameters θ ; (2) measure energy $E(\theta)$ on the quantum device; (3) update classically via $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} E(\theta_t)$; (4) repeat until convergence. The hybrid advantage: the quantum device handles the exponentially hard part (energy computation) while the classical computer handles optimization.

HARDWARE REALITY

Each gate carries $\sim 1\%$ error on current NISQ hardware; circuit depth must remain shallow (20–100 gates) to avoid coherence loss, and **barren plateaus** cause gradients to vanish exponentially in n under random initialization. Fix: use problem-inspired, hardware-efficient ansätze.

4.2 Quantum Approximate Optimization Algorithm (QAOA)

“We show how a quantum computer can efficiently solve combinatorial optimization problems using a time-dependent Hamiltonian that encodes the problem structure.”
— **Farhi, Goldstone & Gutmann (2014)**

THE PROBLEM: MAX-CUT

$$\max_{x \in \{0,1\}^n} \sum_{(i,j) \in E} (1 - x_i x_j)$$

NP-hard. Best classical approximation: Goemans–Williamson, ≈ 0.878 -factor.

QAOA FORMULATION

$$H_P = \sum_{(i,j) \in E} \frac{1}{2}(1 - Z_i Z_j) \quad H_M = \sum_i X_i$$

$$U(\beta, \gamma) = \prod_{p=1}^P e^{-i\beta_p H_M} e^{-i\gamma_p H_P}$$

P = circuit depth (higher $P \rightarrow$ better approximation, harder to optimize). For $p = 1$, QAOA achieves ≈ 0.692 -approximation on random Max-Cut (vs. 0.5 random). For portfolio optimization, the Markowitz problem can be encoded as Max-Weight-Cut and solved with QAOA.

4.3 Quantum Error Correction & Surface Codes

“Error correction need not be done by measuring a large number of qubits. Topological codes can correct arbitrary errors using only local measurements.”
— **Alexei Kitaev (2003)**

THE CHALLENGE

Quantum decoherence: every gate carries error rate $\epsilon \sim 10^{-3}$ to 10^{-4} . Fault-tolerant quantum computing requires quantum error correction.

SURFACE CODE

A 2D grid of physical qubits with stabilizer operators that measure parity without disrupting logical information:

$$S_{\text{plaq}} = Z_1 Z_2 Z_3 Z_4 \quad S_{\text{star}} = X_1 X_2 X_3 X_4$$

Eigenvalue measurements (± 1) yield the syndrome, decoded via $\text{error} = \arg \min_e D(\text{syndrome}, S(e))$.

Key result — the threshold theorem: if physical error rate $\epsilon < \epsilon_{\text{threshold}} \approx 10^{-3}$, the logical error rate decays exponentially with code distance d :

$$\epsilon_L \sim (\epsilon / \epsilon_{\text{th}})^{(d+1)/2}$$

Google Willow (2024) approaches this threshold for the first time; IBM and IonQ remain within a factor of ~ 10 .

◆ SUMMARY: UNIFYING PRINCIPLES

Across All Domains

navy				
Dimensionality Reduction	Shrinkage estimators	LASSO / RIP	LoRA (rank- r)	Shallow circuits
lightgrey Optimization Under Uncertainty	Stochastic control (SDEs)	Causal estimation	Non-convex landscape	Barren plateaus
Information-Theoretic Bounds	Sample complexity · generalization · channel capacity			Measurement complexity

Across all twelve projects, a single mathematical story recurs: **high-dimensional problems become tractable once we exploit low-dimensional structure** — sparsity, low rank, or shallow circuit depth — while managing the fundamental trade-off between statistical and computational efficiency.

Complete Bibliography

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- Björk, T. (2009). *Arbitrage Theory in Continuous Time* (3rd ed.).
- Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*.
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- Peruzzo, A., et al. (2014). *A Variational Eigenvalue Solver on a Photonic Quantum Processor*.

NOBEL PRIZE & TURING AWARD RECOGNITION

- Economics 1990 — Harry Markowitz, Portfolio Theory
- Economics 1994 — John Nash, Non-Cooperative Games
- Economics 1996 — William Vickrey, Mechanism Design
- Economics 1997 — Myron Scholes & Robert Merton, Option Pricing
- Physics 2025 — John Clarke, Michel Devoret & John Martinis, Superconducting Qubits
- Turing Award 2022 — Charles Bennett & Gilles Brassard, Quantum Information Science

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